

Phase 1: User Onboarding

Goals (Multi-select)



- ☒ Reducing carbon footprint
- ☒ Saving energy bills
- ☒ Improving performance
- ☒ Just curious

Persuasion Style (Single-select)



- ☐ Analyst (data-forward)
- ☒ Coach (encouraging)
- ☐ Pragmatist (self-benefit)

Autonomy Constraints



Quiet hours: 21:00 - 09:00

Daily budget $C \in \{0, 1, 2, 3, 4\}$ (user set)

Per-behavior opt-outs

Behavior Scopes



Tabs ✓ Streaming ✓
Printing ✗ Large Transfer ✓
[Lock] All settings stored locally

User-Centered

Phase 2: Adaptive Engine

Context Features



- User: goals, persuasion style
- Behavior: tab count, streaming state
- Device: time, battery%, CPU load
- History: recent votes, dismiss rate

Contextual Bandit

$$\max_{\pi} \mathbb{E} [r(a_t | x_t)] \text{ s.t. } \mathbb{E} [c(a_t | x_t)] \leq C, a_t \notin Q,$$

Feasibility filter:

budget C , quiet hours Q , opt-outs (pre-selection)

$$\mathcal{A} = \{\text{DoNothing}\} \cup \{\text{Targets} \times \text{Styles}\}$$

Targets $\in \{\text{Tabs, Streaming, Printing, Large-Transfer}\}$

Styles $\in \{\text{Analyst, Coach, Pragmatist}\}$

Templates ablation-matched across styles/variants



DoNothing

Key Innovation



[Lock] On-Device Learning



- Ridge regularization; safe exploration (ϵ -decay)
 - Propensity logging; idle-time updates
- ✓ All computation on-device

Privacy-Preserving

Phase 3: Feedback & XAI

Nudge Display

"You have 18 tabs open; closing idle tabs reduces energy use."



You chose Carbon and Analyst; 18 tabs; within your daily budget

User Feedback

Helpful

+1.0

Annoying

-1.0

Wrong time

-0.5

→ Shapes future decisions

Real-time Adaptation



- Temporal alignment improves
- DoNothing increases over time
- Personalized to user patterns

Explainable

Arrow Labeling

---> Preferences (goals, style) ---> Constraints: C (budget), Q (quiet hours), opt-outs ---> Decision (selected template) ---> Feedback: votes → reward update